

Handwritten Log *Digitizer*

A complete workflow for turning paper field notebooks, handwritten datasheets, and analog survey records into clean, structured, analysis-ready digital datasets — without losing a single observation.

BUILT FOR

Field Biologists

Long-Term Monitoring Teams

NGO Data Officers

Graduate Researchers with Paper Records

THE TRANSFORMATION

From "I have years of paper field notebooks that I cannot use in any analysis" to "My handwritten records are a clean, structured, searchable digital dataset ready for R, Python, or any GIS tool."

HOW TO USE THIS PRODUCT

Read Sections 1–3 once. Then go directly to Section 4 (the 6-Step Workflow) and start with your oldest or most critical notebook. Use the AI Prompt Pack (Section 6) at every cleaning step. The Error Taxonomy (Section 8) is your reference when something unexpected appears in the data.

Contents: Core Promise · Pain Points · Desired Outcomes · 6-Step Digitization Workflow · Tool Stack Guide · AI Prompt Pack · 5 Full Notebook-to-Dataset Transformations · Error Taxonomy & Fixes · Column Standardisation Library · Quality Control Checklist · Archive Protocol · Quick Start

SECTION 1

The Core Promise

Somewhere, right now, there are field notebooks from surveys conducted in the 1990s, 2000s, or 2010s that contain irreplaceable biodiversity observations. Some are in institutional archives. Many are in the personal possession of researchers who never had the time or the system to digitize them. A significant portion will never be digitized at all.

This is not a technology problem. Scanning, OCR, and AI tools are better than they have ever been. It is a workflow problem. There is no clear, repeatable process that takes a researcher from a stack of handwritten notebooks to a clean, documented, analysis-ready dataset — without making errors, losing data, or spending six months on what should take six days.

This guide is that workflow. It works for a single notebook from last field season. It works for twenty years of handwritten lemur behavioural observations. It works for data in French, Malagasy, or English. It works whether your handwriting is neat or whether it is the kind of field shorthand only you can read.

THE TRANSFORMATION

Before: "I have years of paper field notebooks I cannot use in any analysis, and I do not know where to start."

After: "My handwritten records are a clean, structured, searchable, and documented digital dataset ready for any analysis tool."

What this system covers: The complete workflow from physical notebook to clean CSV or spreadsheet — including tool selection, scanning protocols, OCR application, AI-assisted transcription and error correction, column standardisation, data validation, quality control, and long-term archiving. Every step is documented, every common error is catalogued, and every stage can be replicated by a second person.

10 Real Pain Points

01 You do not know where to start.

You have a stack of notebooks. Some are organised. Some are not. Some have species codes only you remember. You open the first one and immediately feel overwhelmed. The workflow in this guide removes that paralysis by giving you a decision at each step, not a blank page.

02 OCR fails on field handwriting.

Standard OCR software was trained on printed text. Field notebooks contain abbreviations, crossed-out entries, arrows, side notes, illegible species codes, smudged ink, and condition damage. OCR alone is insufficient — it must be combined with AI interpretation and manual verification. This guide specifies exactly when to use which.

03 You introduce transcription errors while trying to fix the data.

Manual transcription from paper to spreadsheet produces approximately 1 error per 300 keystrokes under normal conditions. Over a large dataset, this is hundreds of corrupted records. The workflow in this guide layers three independent validation steps to catch these before the data is used in any analysis.

04 Your column names and data formats are inconsistent across notebooks.

Season 1 uses "sp_code." Season 2 uses "Species." Season 3 uses "Spp." The date format changes three times. GPS format shifts from degrees-minutes-seconds to decimal degrees midway through a notebook. The Column Standardisation Library in Section 9 resolves all of these systematically.

05 You cannot tell which records are certain and which are questionable.

In the field, observations have varying levels of confidence. A confident species ID written clearly is not the same as a tentative ID written with a question mark and a note about poor light. Without a confidence flag in your dataset, these are indistinguishable from each other.

06 The physical notebooks are deteriorating.

Humidity, insects, mould, and light damage handwritten records faster than most researchers expect. Notebooks from 10–15 years ago in tropical storage conditions are often partially illegible. This guide includes an urgent-first triage system for deteriorating records.

07 You have your own shorthand that nobody else can decode.

Field shorthand is a legitimate and efficient system in the field. In a digital dataset, "Im cf juv 2x rt bank" is not a data record — it is an encoding problem. The workflow guides you through building a personal codebook that translates your shorthand into structured fields.

08 You do not know how to handle genuinely illegible sections.

Every notebook has sections that cannot be read with confidence. The correct response is not to guess and not to omit — it is to flag the record explicitly with an illegibility code, preserve what is readable, and document what is missing. The Error Taxonomy in Section 8 covers every category of illegibility.

09 You have no plan for long-term preservation of the physical notebooks.

Digitizing is not the same as archiving. The workflow in this guide produces both a verified digital dataset and a properly archived physical notebook — with a documented relationship between the two.

10 You cannot verify the accuracy of your own transcription.

Checking your own work is cognitively difficult — your brain reads what it expects to see. This guide includes structured cross-checking protocols that use a second person, an AI tool, or a systematic time-delayed re-read to catch the errors that the transcriber misses.

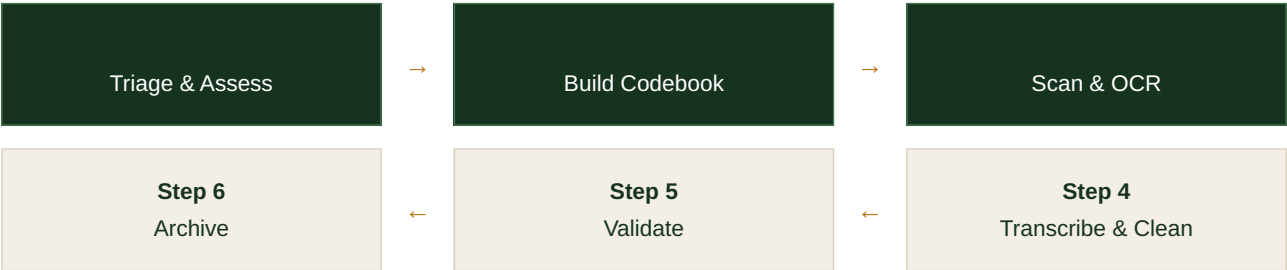
SECTION 3

10 Desired Outcomes

#	Outcome
01	Complete the digitization of one field notebook in a single focused work session of 3–6 hours.
02	Produce a clean, column-standardised CSV or spreadsheet with full provenance documentation.
03	Handle illegible, damaged, and ambiguous records systematically — with no data invented and no data silently lost.
04	Build a personal codebook that translates field shorthand into structured data fields.
05	Merge records from multiple notebooks and multiple seasons into a single, consistent, queryable dataset.
06	Flag confidence levels for every observation so that analyses can filter by certainty.
07	Validate the final dataset against three independent checks before any analysis is run.
08	Archive the physical notebooks and the digital files with a documented, findable relationship between them.
09	Use the digitized dataset in R, Python, QGIS, or any standard analysis tool without additional cleaning.
10	Have a repeatable workflow you can apply to every future notebook — and teach to any team member.

The 6-Step Digitization Workflow

This workflow applies to any handwritten field record — from a single survey form to thirty years of notebooks. Every step produces a documented output. Do not skip steps. The sequence is designed to catch errors at the cheapest possible point in the process.



TRIAGE AND ASSESS

Before digitizing anything, assess the full set of notebooks. This takes 30–60 minutes and determines the order of work and the tools required.

TRIAGE ASSESSMENT QUESTIONS — FOR EACH NOTEBOOK

- Physical condition: Good / Fair (minor damage) / Poor (significant deterioration) / Critical (immediate risk of loss)
- Handwriting legibility: Clear / Mixed / Difficult / Requires specialist knowledge of author's shorthand
- Structure: Does the notebook use a consistent format page-to-page, or is it free-form?
- Language: English / French / Malagasy / Mixed / Code-heavy
- Data types present: Species observations / Transect data / GPS waypoints / Behavioural records / Environmental measurements / Mixed
- Priority: Is this data already used in a publication, or is it unpublished baseline data?

Output of Step 1: A triage log. One row per notebook. Columns: notebook ID, date range, condition, legibility, priority, estimated digitization time.

CRITICAL PRIORITY RULE

Digitize "Critical" condition notebooks FIRST, regardless of scientific priority. Physical deterioration is irreversible. A notebook you cannot read next year is more urgent than a legible notebook from last season.

BUILD THE CODEBOOK

Before transcribing a single observation, document every abbreviation, symbol, and code used in the notebook. This is the most commonly skipped step — and the one that causes the most errors downstream.

CODEBOOK TEMPLATE

For each code, abbreviation, or symbol in the notebook, create one row:

- **Code as written:** exactly as it appears in the notebook (e.g., "Lc", "juv", "rt bank", "x3", "?")
- **Meaning:** the full, unambiguous interpretation (e.g., "Lemur catta", "juvenile", "right river bank", "three individuals", "uncertain identification")
- **Digital field:** which column in the final dataset this maps to (e.g., "species_code", "age_class", "position", "count", "confidence_flag")
- **Digital value:** the standardised value to enter in that field (e.g., "LEMURCATTA", "JUV", "RIGHT_BANK", "3", "UNCERTAIN")

Output of Step 2: A codebook CSV or sheet. Keep it open throughout the entire transcription process.

Use Prompt 1 in the AI Prompt Pack (Section 6) to help build the codebook if you have access to a legible overview of the notebook's contents. The AI can suggest likely interpretations for common field abbreviations — but you must verify every entry against the actual notebook.

SCAN AND OCR

Create a high-resolution digital image of every page before any manual transcription. This is your archival copy and your OCR source. The scan comes before the transcription — always.

SCANNING SPECIFICATIONS

- **Resolution:** 300 DPI minimum for OCR. 600 DPI for damaged or faded notebooks. 1200 DPI for notebooks with very small handwriting or fine detail.
- **Format:** Save as high-quality JPG (for storage efficiency) or TIFF (for maximum quality preservation).
- **Colour mode:** Colour scan, not greyscale — even for pencil notebooks. Colour captures ink degradation, water damage, and annotation colour-coding.
- **Naming convention:** NotebookID_PageNumber_Date.jpg (e.g., NB001_P042_2026-05-01.jpg). Never use spaces in filenames.
- **Equipment:** Flatbed scanner preferred. Document camera acceptable. Phone camera with a scanner app (Microsoft Lens, Adobe Scan) is usable for Good-condition notebooks.

OCR TOOL SELECTION

For printed field forms with handwritten entries: Adobe Acrobat OCR, ABBYY FineReader. These handle structured documents with mixed print/handwriting reasonably well.

For fully handwritten notebooks: Google Document AI, AWS Textract, or Microsoft Azure Document Intelligence. These are cloud-based services with free tiers adequate for most field notebook volumes.

For any notebook: AI-assisted transcription (GPT-4 Vision, Claude) is often more accurate than pure OCR for field handwriting. See Prompt 2. Use this as a complement to OCR, not a replacement for verification.

Output of Step 3: (a) Complete scan archive, one folder per notebook. (b) OCR raw output text file for each page. Do not clean the OCR output yet — preserve it as a reference.

TRANSCRIBE AND CLEAN

The transcription phase produces the first version of the digital dataset. Work page by page, using the codebook (Step 2) and the scanned images (Step 3) simultaneously. Never transcribe from the physical notebook at this stage — always from the scan.

TRANSCRIPTION RULES

- One row per observation. Never combine two observations in one row.
- Use the standardised values from the codebook, not the original shorthand.
- Add a `source_page` column: the scan filename or page number the record was taken from.
- Add a `confidence` column: HIGH (clearly legible, certain identification) / MEDIUM (legible, uncertain ID) / LOW (partially illegible) / ILLEGIBLE (cannot be transcribed).
- Add a `transcriber_note` column: for any record requiring explanation or flagging. Leave blank if no note needed.
- For genuinely illegible records: enter "ILLEGIBLE" in every field that cannot be read. Do not leave blank — blank fields are ambiguous. ILLEGIBLE is unambiguous.

Use Prompts 3 and 4 from the AI Prompt Pack to assist with difficult sections — species name disambiguation, date format correction, and contextual interpretation of ambiguous entries.

Output of Step 4: Raw transcription spreadsheet — unchecked but complete. One row per observation. All columns populated (with ILLEGIBLE where required).

VALIDATE — THREE INDEPENDENT CHECKS

Never skip validation. The three checks are designed to catch different categories of error and must be performed in sequence.

CHECK 1 — AUTOMATED STRUCTURAL VALIDATION

Run the validation prompt (Prompt 7) or use Excel/Google Sheets data validation rules to check: (a) date formats are consistent, (b) GPS coordinates are in the correct range for your study area, (c) species codes are all in the master species list, (d) count fields contain only integers or ILLEGIBLE, (e) no required field is blank.

CHECK 2 — CROSS-REFERENCE AGAINST ORIGINAL SCAN

Take 10% of records at random. Open the scan for each record. Verify that every field in the dataset matches what is written on the page. If error rate exceeds 2%, re-check the entire dataset. If error rate is below 2%, document it and proceed.

CHECK 3 — ECOLOGICAL PLAUSIBILITY REVIEW

Review the dataset for records that are ecologically implausible: a species outside its known range, a count of 500 individuals for a solitary species, a time-of-day entry for a strictly nocturnal species observed at noon. These are not necessarily errors — they may be genuine rare records. Flag them for expert review rather than deleting them.

Output of Step 5: Validated dataset with a validation log documenting all errors found and how they were resolved.

ARCHIVE — PHYSICAL AND DIGITAL

Digitization is not archiving. Both the physical notebook and the digital files require systematic long-term storage with a documented link between them.

PHYSICAL NOTEBOOK ARCHIVING

- Store each notebook in an individual acid-free paper envelope or archival sleeve.
- Label the envelope with: notebook ID, date range, survey area, name of field researcher, digitization date, digital file location.
- Store in a cool, dry, dark location. Avoid cardboard boxes — use archival boxes where possible.
- Do not store rubber-banded or under pressure. Pages must lay flat.

DIGITAL FILE ARCHIVING — THE 3-2-1 RULE

- **3 copies:** The working copy on your computer, a backup on an external drive, and a third copy in cloud storage (Google Drive, Zenodo, institutional repository).
- **2 storage types:** Computer hard drive + external SSD or USB drive.
- **1 offsite:** Cloud storage counts as offsite. An institutional server counts as offsite.

Output of Step 6: Archived physical notebook + archived digital package (scans, codebook, raw transcription, validated dataset, validation log, metadata file).

Tool Stack Guide

Every tool needed for the workflow, organised by function, with cost, platform, and field-specific notes. All free options are indicated.

Microsoft Lens / Adobe Scan

FREE · iOS + Android

Use for	Mobile scanning of Good-condition notebooks. Auto-crops, corrects perspective, adjusts contrast.
Strengths	Fast, free, works offline, produces PDF or JPG. Microsoft Lens integrates with OneDrive.
Limitations	Not sufficient for Critical-condition notebooks. Phone camera resolution varies. Lighting sensitivity.
Field note	Use in landscape mode with the notebook flat. Natural light from a window is better than flash.

Google Document AI / AWS Textract

Free tier available · Cloud-based

Use for	OCR on field handwriting. Significantly more accurate than offline OCR on handwritten text.
Strengths	Handles mixed print/handwriting. Returns structured JSON with confidence scores per word.
Limitations	Requires internet. Data is sent to a cloud provider — check your data sharing agreements before using. Free tier: ~1,000 pages/month.
Alternative	Tesseract OCR (free, offline, open-source) — lower accuracy on handwriting but no data sharing concern.

Claude / GPT-4 Vision (AI transcription assistant)

Free tier / ~\$20/month

Use for	Interpreting difficult handwriting, resolving abbreviations, converting raw OCR output to structured fields.
Strengths	Contextual interpretation. Can use the codebook as context. Handles mixed-language entries. Much better than OCR alone for field shorthand.
Critical limitation	AI transcription must be verified. The AI will make plausible-sounding errors — especially for species names and numerical values. Never use AI output without cross-checking against the scan.
Data privacy	Do not upload sensitive unpublished data (species locations, community partner details) to commercial AI services without consent.

Google Sheets / Microsoft Excel

Free / included in Microsoft 365

Use for	The transcription spreadsheet. Data validation rules. The codebook. The validation log.
Key features	Data Validation (dropdown lists for standardised values), Conditional Formatting (flag outliers), VLOOKUP / XLOOKUP (check species codes against master list).
Field note	Work in Google Sheets if multiple people are transcribing simultaneously. Excel is better for offline work in the field.

R / Python (pandas)

FREE · Cross-platform

Use for	Automated validation checks. Merging multiple notebook datasets. Detecting outliers. Generating the validation report.
Key functions	R: readr, dplyr, validate package. Python: pandas, great_expectations library for data validation.
When needed	For datasets larger than ~2,000 rows, or when merging 5+ notebooks, manual spreadsheet validation becomes impractical. Use R or Python at that scale.

Zenodo / Figshare

FREE · Long-term data repository

Use for	Permanent archiving of digitized datasets. Generates a DOI (digital object identifier) for citability. Required by many funders for data management plans.
Strengths	Free. Persistent. Findable. Generates a DOI. CERN-maintained (Zenodo) — will not disappear.
Access control	Can be set to open access, restricted access (requires permission request), or embargoed (access opens after a set date). Use restricted or embargoed for unpublished data.

AI Prompt Pack — 12 Prompts

These prompts work with Claude, ChatGPT (GPT-4 Vision for image inputs), or any capable AI assistant. Always verify AI output against the original scan. Never use AI output as the final record without cross-checking.

Prompt 1 — Build the Codebook from a Notebook Overview

I am digitizing a handwritten field notebook. I will describe the types of entries it contains. Help me build a codebook that maps field shorthand to standardised digital values. Notebook context: - Survey type: [transect / focal follow / point count / opportunistic observation] - Target taxa: [species or groups recorded] - Language: [English / French / Malagasy / mixed] - Time period: [dates of notebook] - I know the following abbreviations are used: [list any you recognise] For each code type in this notebook (species codes, behavioural codes, condition codes, habitat codes, equipment codes), suggest: 1. Likely meanings based on context 2. Recommended standardised column name 3. Recommended standardised digital value (e.g., all caps, underscores, no spaces) I will verify every suggestion against the actual notebook.

Prompt 2 — Transcribe a Handwritten Page (Image Input)

I am attaching a scan of a handwritten field notebook page. Please transcribe every entry on this page into a structured table. Context: - Survey type: [describe] - Target taxa: [describe] - Known abbreviations: [paste your codebook here] - Column structure expected: [date | time | species | count | age | sex | behaviour | GPS | habitat | observer | notes | confidence] For each entry: - Transcribe exactly what is written in the "raw" column - Interpret using the codebook in the "interpreted" column - Rate your confidence: HIGH (clearly legible) / MEDIUM (interpretable with context) / LOW (partially illegible) / ILLEGIBLE - Add a "transcriber_note" for any ambiguity Do NOT invent values. If you cannot read a field, write ILLEGIBLE. [ATTACH SCAN IMAGE]

Prompt 3 — Resolve Ambiguous Species Names

I am transcribing a field notebook from [country/region] recording [target taxa]. I have the following ambiguous entries that may be species names or abbreviations: [LIST THE AMBIGUOUS ENTRIES, ONE PER LINE] For each entry: 1. Suggest the most likely intended species, given the geographic context 2. List alternative interpretations if more than one is plausible 3. Recommend whether to flag as HIGH, MEDIUM, or LOW confidence 4. Suggest a note for the transcriber_note column I will verify your suggestions against the original notebook and a regional species checklist. Geographic region: [country and habitat type] Survey period: [dates] Observer expertise level: [expert herpetologist / general field biologist / student]

Prompt 4 — Standardise Inconsistent Date Formats

I have a column of dates from a digitized field notebook that contains multiple inconsistent formats. Please standardise all dates to ISO 8601 format (YYYY-MM-DD). Rules: - If the year is ambiguous (e.g., "15" could be 2015 or 1915), use [specify the likely year range based on notebook context] - If only month and day are present (no year), flag as INCOMPLETE_DATE and add the notebook year in a separate "assumed_year" column - If the date is completely illegible, return ILLEGIBLE - Do NOT invent dates – flag ambiguities, do not resolve them by guessing Date column to standardise: [PASTE YOUR DATE COLUMN HERE – one date per line]

Prompt 5 — Clean and Standardise GPS Coordinates

I have GPS coordinates from a handwritten field notebook in inconsistent formats. Please standardise all coordinates to decimal degrees (DD.DDDDD format) in WGS84. Rules: - Convert degrees-minutes-seconds (DMS) to decimal degrees - Convert degrees-decimal minutes (DDM) to decimal degrees - If hemisphere is missing, infer from the study area context: [specify latitude and longitude ranges for your study area] - If coordinates are clearly outside the study area range [specify bounding box], flag as SPATIAL_OUTLIER – do not correct - If coordinates are illegible or missing, return ILLEGIBLE Coordinate column to clean: [PASTE YOUR COORDINATE COLUMN HERE – one coordinate per line] Study area bounding box: Latitude: [min] to [max] Longitude: [min] to [max]

Prompt 6 — Interpret Behavioural Observation Codes

I am digitizing focal follow behavioural records for [species name]. The notebook uses behavioural codes I need to interpret. Below is a list of codes as they appear in the notebook. Known context: - Species: [name] - Ethogram used: [if known – e.g., standard lemur ethogram, custom ethogram] - Observer: [field researcher name or description] - Survey period: [dates] For each code: 1. Suggest the most likely behavioural category (locomotion / feeding / social / rest / vigilance / other) 2. Suggest the standardised behaviour description 3. Flag confidence level Codes to interpret: [LIST CODES, ONE PER LINE]

Prompt 7 — Automated Structural Validation

I have a digitized field dataset. Please validate it against the following rules and report all violations. Dataset structure: [describe columns and expected formats] Validation rules to apply: 1. Date format: all dates must match YYYY-MM-DD 2. GPS coordinates: latitude between [min] and [max], longitude between [min] and [max] 3. Species codes: must appear in this master list: [paste species code list] 4. Count field: must be a positive integer or ILLEGIBLE – no negatives, no decimals, no text 5. Confidence field: must be one of: HIGH / MEDIUM / LOW / ILLEGIBLE 6. No field in the following required columns may be blank: [list required columns] 7. Observer codes must appear in this list: [paste observer code list] For each violation: row number, column, the value found, the rule violated, recommended action. Dataset to validate (paste as CSV): [PASTE DATASET HERE]

Prompt 8 — Merge Multiple Notebook Datasets

I have [number] digitized notebook datasets from different field seasons that I need to merge into a single dataset. Each dataset has slightly different column names and formats. I will describe each dataset below. Please generate: 1. A unified column schema (standardised name, format, description) that accommodates all datasets 2. A mapping table showing how each original column name maps to the unified name 3. A list of values that need to be standardised across datasets (e.g., "Lc" in dataset 1 = "LEMURCATTA" in dataset 3) 4. Flags for any columns present in some datasets but not others – recommend whether to include with NAs or exclude Dataset descriptions: [Describe each dataset: column names, date range, survey type, any known inconsistencies]

Prompt 9 — Generate Dataset Metadata File

I have completed digitizing a field notebook dataset. Please help me write a metadata file that documents this dataset. Dataset details: - Original notebook: [ID, date range, observer, survey site] - Data types: [species observations / transect data / behavioural records / etc.] - Geographic coverage: [study area name and bounding coordinates] - Taxonomic coverage: [taxa recorded] - Total records: [number] - Records flagged as ILLEGIBLE: [number and percentage] - Digitization date: [date] - Digitizer: [name] - Validation completed: [yes/no, validation method used] - Known gaps or limitations: [describe any sections not digitized, damaged pages, etc.] - Codebook reference: [filename] - Scan archive location: [folder path or URL] Format: Darwin Core compatible metadata, suitable for upload to GBIF or a data repository.

Prompt 10 — Prioritise Illegible Records for Expert Review

I have a digitized dataset with [number] records flagged as LOW confidence or ILLEGIBLE. I need to prioritise which records to send for expert review. Please analyse the following illegible/uncertain records and rank them by: 1. Scientific importance (rare species, unusual location, rare behaviour) 2. Recoverability (how likely is it that an expert could interpret the original scan?) 3. Impact on analysis (would this record significantly affect diversity counts, distribution maps, or population estimates?) For each record, recommend: PRIORITISE FOR EXPERT REVIEW / ACCEPT CURRENT TRANSCRIPTION / ACCEPT AS ILLEGIBLE Dataset of uncertain records (paste as CSV): [PASTE UNCERTAIN RECORDS HERE] Context: - Study area and target taxa: [describe] - Primary analysis objectives: [what the data will be used for]

Prompt 11 — Write a Data Quality Report

I have completed digitizing and validating a field notebook dataset. Please write a short data quality report based on the following information. Dataset summary: - Total records: [number] - Records with HIGH confidence: [number / percentage] - Records with MEDIUM confidence: [number / percentage] - Records with LOW confidence: [number / percentage] - Records flagged ILLEGIBLE: [number / percentage] - Validation errors found and resolved: [describe types and counts] - Validation errors remaining (expert review pending): [number] - Temporal coverage: [date range] - Spatial coverage: [study area, number of sites] The report should: - State the overall usability of the dataset for analysis - Specify which record subsets are suitable for which types of analysis (e.g., HIGH confidence records only for species distribution modelling; all records for presence/absence summaries) - Note any known limitations that future users should understand - Recommend whether expert review of LOW confidence records should precede or follow the primary analysis - Format: 200–300 words, professional research data quality report tone

Prompt 12 — French Language Notebook Transcription

I am digitizing a handwritten field notebook written primarily in French, with some Malagasy terms and standard Latin species names. Please transcribe the attached page and:

- Transcribe French entries in their original form in the "raw_french" column - Provide an English translation in the "translated_english" column - Preserve all Latin species names unchanged - For Malagasy terms (habitat names, local species names, place names): transcribe as written, add a note that these require local expert verification - Flag any entry where the French is ambiguous or uses specialist terminology that affects meaning

Known context: - Survey type: [describe] - Field researcher: [description – e.g., Malagasy biologist trained in French] - Survey region: [region of Madagascar] [ATTACH SCAN IMAGE]

5 Full Notebook-to-Dataset Transformations

Complete worked examples from raw handwritten entry to clean digital record. Each shows the original notation, the interpretation process, and the final dataset row.

Transformation 1 — Lemur Focal Follow Record

RAW NOTEBOOK ENTRY — AS WRITTEN

06/03/24 0742 Lc ♀ad "Maya" rt bank fig tree ~4m feed figs 12min → mov W 0754 Lc juv
x2 base same tree play 6min 0800 all 3 mov W out of sight alrt call bef dep

↓ Codebook applied + structured into rows ↓

date	time_start	species	age_sex	individual_id	count	behaviour	habitat
2024-03-06	07:42	LEMURCATTA	F_AD	Maya	1	FEED	FIG_TREE_4M
2024-03-06	07:54	LEMURCATTA	JUV	UNKNOWN	2	PLAY	FIG_TREE_BASE
2024-03-06	08:00	LEMURCATTA	MIXED	UNKNOWN	3	MOVE	UNKNOWN

DECISIONS MADE DURING THIS TRANSFORMATION

The "~4m" was recorded as a habitat height estimate, not a precise measurement — placed in notes, not a numeric field. "alrt call bef dep" was decoded from context as alert call before departure. The approximate sign (~) was preserved in the notes column but the height estimate was not entered as a numeric value due to uncertainty. Duration for the third row was not stated in the notebook — entered as ILLEGIBLE, not inferred.

Transformation 2 — Herpetofauna Transect Record

RAW NOTEBOOK ENTRY — AS WRITTEN

T3-N · 23:15 · stream edge · Mantellasp? small red · 1 · coord: S18°27.456 E47°11.234
tissue→vial7 · ill. light conditions · ID uncertain cf. aurantiaca

↓ Codebook + GPS conversion + uncertainty flagging ↓

transect_id	time	habitat	species_raw	species_interpreted	count	lat_dd	lon_dd
T3-N	23:15	STREAM_EDGE	Mantellasp? small red	Mantella cf. aurantiaca	1	-18.45760	47.18723

GPS CONVERSION NOTE

S18°27.456 / E47°11.234 is in degrees-decimal minutes format. Converted: latitude = $-(18 + 27.456/60) = -18.45760$. Longitude = $47 + 11.234/60 = 47.18723$. This conversion must be shown in the validation log — do not silently convert without documenting the original format and the conversion formula applied.

Transformation 3 — Camera Trap Check Log

RAW NOTEBOOK ENTRY — AS WRITTEN

CT-12 checked 15/4 · 847 triggers · card swapped (old→bag3/card12) · batt ok ·
images: fosa x3 noc, Lemcatta grp 7+ dawn, lots of birds daytime, 1 × human 0300(!)
camera slightly displaced – reset 45cm / perpendic path – locking cable intact

↓ Structured into camera trap check dataset ↓

camera_id	check_date	triggers_total	card_in	card_out	battery_status	species_detected	
CT-12	2024-04-15	847	BAG3_CARD12	ILLEGIBLE	GOOD	Cryptoprocta ferox; Lemur catta; UNID_BIRD; UNID_HUMAN	D

DECISIONS MADE

"Old" card identity was not recorded in the notebook — entered as ILLEGIBLE, not guessed. Human detection flagged separately in notes for security/ethics review. Species group count "7+" recorded as a text note, not a numeric field, because the "+" indicates the count was uncertain.

Transformation 4 — Drone Survey Flight Log

RAW NOTEBOOK ENTRY — AS WRITTEN

26 Jul · Mavic3T · Ihofa NE · dep 0612 land 0637 · alt 80m · 3 thermal passes wind ~15kph NW · humid · batt1=87% start 12% end · batt2=91% start 8% end detections: thermal sig ×11 canopy level →3 confirmed Indri (calls heard sim.) remaining 8 unconfirmed — too small for Indri, poss. bird or sm mammal card: DJIM3T_2026-07-26_001 · pilot: HFR · conditions: acceptable

↓ Structured into drone survey log dataset ↓

flight_date	aircraft	site	depart	land	alt_m	passes	wind_kph	detections_total
2026-07-26	DJI_MAVIC3T	IHOFA_NE	06:12	06:37	80	3	15	11

Transformation 5 — Multi-season, Partially Illegible Vegetation Plot Record

RAW NOTEBOOK ENTRY — AS WRITTEN (DETERIORATED NOTEBOOK, 2009)

Plot F[?]4 · [smudge] 2009 · 10×10m · canopy cov est [illegible]% · litter depth 4.5cm Trees DBH>10: Dalb[illegible] sp ×3 (DBH 12,15,[?]cm) · Grewia sp ×1 (DBH 23cm) understory: dense · access: diff

↓ Preserved illegibilities, no invented values ↓

plot_id	survey_date	plot_size_m2	canopy_cover_pct	litter_depth_cm	species	count
F_ILLEGIBLE_4	ILLEGIBLE_2009	100	ILLEGIBLE	4.5	Dalbergia sp.	3
F_ILLEGIBLE_4	ILLEGIBLE_2009	100	ILLEGIBLE	4.5	Grewia sp.	1

CRITICAL PRINCIPLE DEMONSTRATED HERE

The plot ID was ambiguous between F4 and F14. The correct action was to enter "F_ILLEGIBLE_4" and document the ambiguity in notes — not to guess F4 or F14. If this data is later linked to spatial records, the ambiguity can potentially be resolved using GPS coordinates or cross-referencing with other notebooks from the same survey. A guessed value would destroy that possibility permanently.

Error Taxonomy and Fixes

Every category of error encountered in handwritten field data digitization, with the correct response for each. Use this as a reference when something unexpected appears during transcription or validation.

Error type	Example	Correct response	Do NOT do this
Illegible field — complete	Water damage makes an entire cell unreadable	Enter ILLEGIBLE in the field. Document in transcriber_note. Confidence = ILLEGIBLE.	Leave blank. Guess. Interpolate from surrounding records.
Illegible field — partial	"Grewia s[smudge]" — species partly visible	Enter what is legible + ILLEGIBLE for the rest: "Grewia sp. ILLEGIBLE". Note in transcriber_note.	Complete the name by guessing the missing characters.
Ambiguous abbreviation	"Lf" — could be Lemur fulvus or Leptosomus flavicans	Enter "LF_AMBIGUOUS" in the field. List both interpretations in transcriber_note. Confidence = LOW.	Choose one interpretation without documenting the ambiguity.
Inconsistent date format	3/4/24 vs 04.03.2024 in the same notebook	Standardise all dates to YYYY-MM-DD. Document both original formats in the codebook.	Leave dates in mixed formats in the final dataset.
Missing required field	Time of observation not recorded on a page	Enter MISSING (not blank). Document in notes. If time can be inferred from surrounding records, enter as INFERRED_HH:MM and note the inference basis.	Leave blank. Infer silently without flagging.
Crossed-out entry	An observation is struck through in the notebook	Include the record with a "VOIDED" confidence flag and a note: "crossed out in original notebook." Do not omit — voided records are evidence.	Silently exclude crossed-out entries.
Contradictory entries	Same animal recorded as adult on line 3 and juvenile on line 5	Include both records exactly as written. Add a transcriber_note flagging the contradiction. Do not resolve it — that is for the researcher.	Choose one value and discard the other.
GPS outside study area	A coordinate that falls in the ocean instead of the forest	Enter the coordinate as written with a SPATIAL_OUTLIER flag. Do not correct. Note likely causes: datum error, transposition, transcription error in the field.	Delete or "correct" the coordinate.
Count as a range	"8-12 individuals observed"	Enter min and max in separate columns: count_min = 8, count_max = 12. Note in transcriber_note.	Average the range and enter "10".

Observer added a later annotation	A later note in different ink adds "ID confirmed by X"	Include in a separate "post_survey_annotation" column with the content of the annotation verbatim.	Incorporate the annotation into the original record without distinguishing it.
Record on wrong page	An entry clearly belongs to a different date based on context	Transcribe as written. Add a note: "Record on date X page but context suggests date Y — do not correct." Flag for researcher review.	Silently move the record to what seems like the correct date.
Duplicate record	The same observation appears twice on different pages	Include both records. Add "POSSIBLE_DUPLICATE" flag to both. Note in transcriber_note: "identical/very similar record also on page X."	Silently delete one of the duplicates.

Column Standardisation Library

Standard column names and format specifications for the most common field biology data types. Use these consistently across all notebooks to enable dataset merging.

Universal Columns — Include in Every Dataset

Column name	Format	Example	Notes
record_id	Text, unique	NB003_P042_R007	Notebook ID + page + row number. Must be globally unique across all notebooks.
source_notebook	Text	NB003	Notebook identifier from triage log.
source_page	Integer	42	Page number in the physical notebook.
scan_filename	Text	NB003_P042_2026-05-01.jpg	Exact filename of the scan used for transcription.
date	ISO 8601: YYYY-MM-DD	2024-03-06	Date of the observation. ILLEGIBLE if not readable.
time	24h: HH:MM	07:42	Time of observation start. MISSING if not recorded.
observer	Text code	HFR	Observer initials or code from codebook.
confidence	Controlled vocabulary	HIGH / MEDIUM / LOW / ILLEGIBLE	Transcription confidence for this record.
transcriber_note	Free text	Species ID uncertain — cf. only	Leave blank if no issues. Document every ambiguity.
digitization_date	ISO 8601: YYYY-MM-DD	2026-05-01	Date this record was transcribed.
digitizer	Text	HFR	Person who transcribed this record.

Species Observation Columns

Column name	Format	Example	Notes
species_raw	Text — exact as written	Lc	The abbreviation or name exactly as written in the notebook.
species_code	Text — standardised	LEMURCATTA	From codebook. ALLCAPS, no spaces, no special characters.
species_latin	Text — binomial	Lemur catta	Full Latin name. "sp." if only genus known. "cf." if uncertain.
count	Integer or range	3	Number of individuals. Use count_min / count_max for ranges.
age_class	AD / JUV / INF / UNKNOWN	AD	Adult / Juvenile / Infant / Unknown.
sex	M / F / UNKNOWN	F	Male / Female / Unknown.
individual_id	Text or UNKNOWN	Maya	Named individual or collar ID if known.

Spatial Columns

Column name	Format	Example	Notes
latitude	Decimal degrees, 5 d.p., WGS84	-18.45760	Negative for South. ILLEGIBLE if unreadable.
longitude	Decimal degrees, 5 d.p., WGS84	47.18723	Positive for East. ILLEGIBLE if unreadable.
coord_original	Text — exact as written	S18°27.456 E47°11.234	Always preserve the original format alongside converted decimal degrees.
spatial_flag	Controlled vocabulary	SPATIAL_OUTLIER	VERIFIED / SPATIAL_OUTLIER / INFERRED / MISSING / ILLEGIBLE
site_name	Text	IHOFA_NE	Named site from codebook. More stable than coordinates for filtering.
transect_id	Text	T3-N	Transect or plot identifier.

Quality Control Checklist

Run this checklist at the end of every notebook digitization before the dataset is used in any analysis.

Completeness

- ☐ Every page of the notebook has been scanned. Page count in scan archive matches page count in physical notebook.
- ☐ Every observation on every page has a corresponding row in the dataset. No observations have been silently omitted.
- ☐ Every illegible or uncertain field is explicitly marked (ILLEGIBLE, MISSING, LOW confidence) — not left blank.
- ☐ The codebook covers every abbreviation that appears in the notebook.

Structural Integrity

- ☐ All date fields are in YYYY-MM-DD format.
- ☐ All time fields are in HH:MM 24-hour format.
- ☐ All GPS coordinates are in decimal degrees, WGS84, with 5 decimal places (or flagged as ILLEGIBLE/MISSING).
- ☐ All species codes appear in the master species list or are flagged as AMBIGUOUS.
- ☐ All confidence values are from the controlled vocabulary: HIGH / MEDIUM / LOW / ILLEGIBLE.
- ☐ No required column contains a blank cell (ILLEGIBLE or MISSING are acceptable — blank is not).
- ☐ record_id values are unique across the entire dataset.

Ecological Plausibility

- ☐ All GPS coordinates fall within the bounding box of the study area (or are flagged as SPATIAL_OUTLIER).
- ☐ Species recorded are within known ranges (or are flagged as RANGE_OUTLIER with a note).
- ☐ Count values are plausible for the species and survey method (or are flagged and noted).

- ☐ Time values are plausible for the species (nocturnal species at night, diurnal at day — outliers are flagged).

Cross-Reference Validation

- ☐ 10% random sample cross-checked against original scans. Error rate documented in validation log.
- ☐ Any error rate above 2% triggers a full re-check of the entire dataset.
- ☐ Validation log is complete: lists every error found, how it was classified, and how it was resolved.

Archive Readiness

- ☐ Scan archive is complete and all files are named according to the naming convention.
- ☐ Codebook file is finalised and saved alongside the dataset.
- ☐ Metadata file is complete (see Prompt 9 in Section 6).
- ☐ Physical notebook is ready for archival storage (acid-free envelope, labelled correctly).
- ☐ 3-2-1 backup completed for the digital package.

SECTION 11

Archive Protocol

Physical Notebook Archive Standard

NOTEBOOK ARCHIVE LABEL – paste or write on the envelope/sleeve Notebook ID: NB___ Date range: [DD/MM/YYYY] to [DD/MM/YYYY] Observer(s): [names] Survey site(s): [site names] Survey type: [transect / focal follow / camera trap / mixed] Physical condition: Good / Fair / Poor / Critical Digitization status: COMPLETE / PARTIAL / NOT YET DIGITIZED Digitization date: [YYYY-MM-DD] Digital file location: [folder path or URL] Dataset ID: [dataset filename] Archived by: [name] Archive date: [YYYY-MM-DD]

Digital Archive Package — One Folder Per Notebook

```
NB003_IH0FA_2024/ ├── scans/ | ├── NB003_P001_2026-05-01.jpg | ├──
NB003_P002_2026-05-01.jpg | └─ ... (one file per page) ├── NB003_codebook.csv ├──
NB003_raw_transcription.csv (unvalidated – preserved for reference) ├──
NB003_validated_dataset.csv (the final, validated dataset) ├── NB003_validation_log.csv
(all errors found and resolutions) ├── NB003_metadata.txt (Darwin Core compatible metadata)
└─ NB003_README.txt (plain language description for future users)
```


Quick Start — Your First Notebook in 6 Hours

Hour 1 — Triage and Setup

- ☐ Lay out all notebooks. Complete a triage assessment row for each one.
 - ☐ Identify your most urgent notebook (Critical condition) or your highest-priority notebook (most needed for current analysis).
 - ☐ Set up your folder structure for the digital archive package.
 - ☐ Open a new spreadsheet. Create the column headers from the Column Standardisation Library (Section 9) for your data type.
-

Hour 2 — Scan and Codebook

- ☐ Scan all pages of the selected notebook. Name files according to the naming convention.
 - ☐ Browse through the scans to identify all abbreviations, codes, and symbols.
 - ☐ Use Prompt 1 to help draft the codebook. Verify every suggestion against the scans.
 - ☐ Save the codebook as a separate CSV file.
-

Hours 3–4 — Transcription

- ☐ Work page by page from the scans (not the physical notebook).
 - ☐ Use Prompt 2 for any difficult pages — attach the scan image to the AI.
 - ☐ Fill every field for every observation. Use ILLEGIBLE or MISSING — never blank.
 - ☐ Use Prompt 3 for ambiguous species names. Use Prompt 4 for date standardisation.
-

Hour 5 — Validation

- ☐ Run Prompt 7 on the complete dataset for automated structural checks.
 - ☐ Cross-check 10% of records against scans. Document the error rate.
-

- ☐ Review all LOW and ILLEGIBLE confidence records for ecological plausibility.
-

- ☐ Complete the validation log.
-

Hour 6 — Archive

- ☐ Generate the metadata file using Prompt 9.
-

- ☐ Label the physical notebook envelope and store it correctly.
-

- ☐ Complete the 3-2-1 digital backup.
-

- ☐ Run the Quality Control Checklist from Section 10.
-

- ☐ Add a row to your triage log: this notebook is now COMPLETE.
-

Minimum Viable vs. Premium Version

Feature	This Version (MVP)	Premium (Future)
6-step digitization workflow	✓	✓
Tool stack guide (6 tools)	✓	✓ Expanded to 12 tools
AI prompt pack (12 prompts)	✓	✓ 20 prompts
5 full notebook-to-dataset transformations	✓	✓ 10 transformations
Error taxonomy (12 categories)	✓	✓ 20 categories
Column standardisation library	✓	✓ Expanded to 8 data types
Quality control checklist	✓	✓
Archive protocol + label templates	✓	✓
R/Python validation scripts	—	Planned
Google Sheets pre-built template with validation rules	—	Planned
Malagasy/French notebook transcription guide	—	Planned
Batch processing workflow for 20+ notebooks	—	Planned
GBIF / Zenodo upload guide	—	Planned

The data in your notebooks already exists. *It just cannot be found yet.*

Every year of handwritten records you rescue from a deteriorating notebook is a year of biodiversity data that re-enters science. The workflow is not glamorous. The result — a clean, documented, citable dataset from work that would otherwise be lost — absolutely is.

Handwritten Log Digitizer v1.0

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